

ThoughtFlow-FP8: A Mixed Gate for Phase-Aware KV Retention

Anonymous authors

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Abstract

ThoughtFlow-FP8 tests whether explicit anchor/phase retention plus byte-aware FP8-style budgeting can preserve useful reasoning context better than matched-budget KV-retention proxies. Synthetic traces support the intuition that phase markers and anchors can be protected, but real retained-context quality weakens the branch. On 24 distilgpt2 traces at a 0.20 keep fraction, a bounded successor policy, ThoughtFlow-saliency-recent, improves over the original phase-only policy and beats LongFlow-like and ThinKV-like retained-text proxies, but narrowly loses to an R-KV-like proxy on continuation NLL (3.434 vs 3.419). We then ran the closest Mac-local cache-quality gate: prune the actual prefix KV cache on CPU and score the continuation from the sparse cache. ThoughtFlow-saliency-recent is the best compressed row in mean NLL on all 24 traces (3.372 vs 3.389 for ThinKV-like and 3.438 for R-KV-like), but the all-trace margin over ThinKV-like is only 0.017 NLL. A new 24-config train-fixed sparse sweep selects a lighter-anchor policy that clears the held-out mean margin versus ThinKV-like and R-KV-like, but its paired CI still crosses zero versus ThinKV-like. We therefore present this as a mixed workshop gate, not a positive method.

1 Policy Family

The original policy protects anchors and phase labels. The first successor adds a small recent-token reserve. The second successor protects anchors, reserves half the retained budget for recent tokens, and lets phase markers compete with math-state and high-importance reasoning tokens under a saliency bonus. This tests whether the failure was caused by over-preserving interpretable markers that do not matter for immediate continuation likelihood.

Policy	Keep rate	NLL	Δ NLL vs full	PPL
full context	1.000	2.101	0.000	9.7
R-KV-like	0.210	3.419	1.319	38.7
ThinKV-like	0.210	3.583	1.482	44.6
LongFlow-like	0.210	3.961	1.861	74.2
ThoughtFlow	0.210	3.961	1.861	74.2
ThoughtFlow-recent	0.210	3.562	1.461	44.2
ThoughtFlow-saliency-recent	0.210	3.434	1.333	38.1
Held-out R-KV-like	0.211	3.482	–	43.7
Held-out sweep best	0.211	3.480	–	43.7

Table 1: Retained-context continuation proxy on distilgpt2 traces. Lower NLL is better. Full context is not a matched-budget baseline. The held-out sweep row is selected on 12 train traces and reported on 12 held-out traces, so its NLL is not directly averaged with the all-trace rows.

2 Hidden/KV Telemetry

We then ran a Mac-local telemetry gate using real distilgpt2 tensors rather than only text markers. The probe ranks retained tokens by attention received, final-hidden norm, key norm, value norm, or combined KV norm. This is still not sparse-KV decoding, but it checks whether the interpretable phase policy is merely preserving labels that hidden/KV saliency would not select.

Policy	Keep rate	Anchor recall	Phase recall	Math-state recall
ThoughtFlow	0.207	1.000	1.000	0.918
LongFlow-like	0.207	1.000	1.000	0.918
ThinKV-like	0.207	1.000	0.948	0.848
value_norm_topk	0.207	0.000	0.492	0.845
train-selected ThoughtFlow sweep	0.207	1.000	0.430	0.956

Table 2: Hidden/KV saliency telemetry on 24 distilgpt2 traces. The strongest real-saliency proxy by phase+math recall is value_norm_topk. ThoughtFlow beats it on phase recall, but LongFlow-like ties ThoughtFlow under the local importance heuristic.

Paired against value_norm_topk, ThoughtFlow’s phase-recall delta is +0.508 with 95% CI [+0.436, +0.579]. The math-state delta is only +0.073 with 95% CI [-0.078, +0.223]. This rules out a reviewer-facing claim based on phase recall alone: it may be marker preservation rather than useful cache retention.

3 CPU Sparse-Cache Quality Gate

Finally, we ran a CPU-only sparse-cache probe. The model processes the full prefix once, each policy prunes the returned KV cache to the same 0.20 budget, and the continuation is scored from the pruned cache. This is not a Triton, CUDA, FP8, latency, or throughput result, but it is stronger than retained-text NLL because the model actually receives a sparse cache.

Policy	Keep rate	NLL	Paired Δ vs R-KV-like
full cache	1.000	2.142	-1.296 [-1.533,-1.066]
ThoughtFlow-saliency-recent	0.210	3.372	-0.067 [-0.151,+0.011]
ThinKV-like	0.210	3.389	-0.049 [-0.192,+0.077]
ThoughtFlow-recent	0.210	3.399	-0.040 [-0.169,+0.074]
ThoughtFlow sweep best	0.210	3.436	-0.002 [-0.007,+0.000]
R-KV-like	0.210	3.438	0.000
LongFlow-like	0.210	3.588	+0.150 [-0.016,+0.300]
ThoughtFlow	0.210	3.588	+0.150 [-0.006,+0.297]

Table 3: CPU sparse-KV continuation quality on 24 distilgpt2 traces. Negative paired delta is better than R-KV-like. ThoughtFlow-saliency-recent is best in mean NLL, but it clears the strongest non-ThoughtFlow row by only 0.017 NLL and remains below the pre-registered 0.03 promotion margin.

A bounded sparse-cache sweep over 24 nearby anchor/recent/phase/math configurations selects tf_sparse_r0.55_p0.05_m0.12_a2 on the 12 train traces. On the 12 held-out traces it gets NLL 3.340 versus ThinKV-like 3.385 and R-KV-like 3.420. This clears the mean 0.03 margin, and the paired delta versus R-KV-like is -0.080 with 95% CI [-0.152,-0.014]. However, the paired delta versus ThinKV-like is -0.045 with 95% CI [-0.226,+0.182]. The fixed ThoughtFlow-saliency-recent incumbent has lower held-out mean NLL, 3.304, but its paired CIs also cross zero. This is a promising falsification candidate, not a positive result.

4 Decision

The branch is mixed but not revived as a positive method. ThoughtFlow-saliency-recent repairs most of the phase-only failure and is now the best mean row under actual CPU cache dropping, and the bounded train-fixed sparse sweep finds a held-out mean win. The blocker is uncertainty against ThinKV-like, the strongest non-ThoughtFlow held-out baseline. A paper-ready claim would require the frozen candidates to beat R-KV-like, LongFlow-like, and ThinKV-like rows at matched bytes on a larger sparse-KV slice with paired uncertainty.

5 Artifacts

- `experimental/thoughtflow_fp8/phase2/perplexity_impact_proxy.md`
- `experimental/thoughtflow_fp8/phase2/policy_sweep.md`
- `experimental/thoughtflow_fp8/phase2/hidden_saliency_retention_probe.md`
- `experimental/thoughtflow_fp8/phase2/kv_drop_quality_probe.md`
- `experimental/thoughtflow_fp8/phase2/real_trace_retention_sweep.md`
- `experimental/thoughtflow_fp8/paper/colm_workshop_shell.md`